

ENCARNACION, Paraguay

WMO: 862970

Lat: 27.317S

Lon: 55.833W

Elev: 299

StdP: 14.54

Time Zone: -4.00 (PRY)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	38.0	41.9	32.2	26.9	45.6	35.3	30.6	48.3	23.5	73.4	19.7	71.0	2.2	50	0.341

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.0	96.1	75.2	93.9	74.6	91.8	74.8	79.7	89.7	78.6	88.4	77.6	87.1	10.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	141.7	85.6	75.8	136.4	84.4	74.7	131.6	83.3	43.6	90.7	42.4	87.6	41.4	88.6	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	17.6	15.9		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.3	80.3	78.7	76.8	71.6	66.4	62.2	62.1	64.4	68.0	72.9	74.1	78.8
	DBStd	9.38	3.89	4.13	4.46	6.65	7.49	8.99	9.84	9.26	8.33	6.10	5.28	4.40
	HDD50	31	0	0	0	0	1	10	15	4	1	0	0	0
	HDD65	633	0	0	0	20	76	159	180	130	60	5	3	0
	CDD50	7816	940	804	830	648	508	377	389	451	542	709	724	894
	CDD65	2941	475	384	365	218	118	76	89	111	151	249	276	429
	CDH74	26191	4921	3506	3124	1621	689	373	518	934	1336	2173	2584	4412
	CDH80	10425	2135	1429	1232	551	180	63	113	325	548	883	995	1971
Wind	WSAvg	6.9	6.3	6.0	6.1	6.4	6.6	7.2	7.2	7.3	7.5	7.9	7.3	6.8
Precipitation	PrecAvg	70.6	6.1	6.0	5.6	6.9	5.5	5.2	3.7	4.0	5.4	7.8	6.6	6.9
	PrecMax	112.1	12.9	18.9	16.6	17.8	22.9	14.9	13.9	11.9	14.9	18.6	21.9	21.1
	PrecMin	43.3	0.4	0.3	0.5	0.0	0.4	0.3	0.2	0.2	0.7	0.9	0.1	0.4
	PrecStd	16.9	3.5	3.7	3.2	4.6	3.9	3.2	2.6	2.6	3.0	4.2	4.6	4.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.5	96.1	94.5	91.8	89.1	84.6	86.5	91.8	95.9	98.6	96.3	97.8
		MCWB	76.7	75.0	74.9	74.9	72.9	69.7	70.0	70.5	71.7	75.1	73.8	75.1
	2%	DB	94.8	93.2	91.8	89.4	84.8	81.7	83.0	87.6	91.3	92.9	92.5	95.3
		MCWB	76.9	75.1	76.2	74.6	71.5	68.7	68.3	69.4	70.9	74.6	71.9	74.5
	5%	DB	92.4	90.8	89.6	86.6	81.3	78.7	80.2	83.4	86.9	88.2	89.4	92.8
		MCWB	76.5	75.4	75.0	73.6	69.8	67.9	68.2	68.1	69.1	72.9	71.4	74.9
	10%	DB	89.9	88.3	87.0	83.4	77.5	75.3	76.4	79.1	81.8	84.4	86.0	89.6
		MCWB	75.9	74.9	74.1	72.2	69.4	66.3	66.5	66.7	66.9	71.3	70.1	74.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.1	80.4	80.1	77.9	74.7	73.3	74.2	74.2	74.5	78.7	78.0	80.6
		MCDB	91.6	88.9	90.2	87.4	83.6	80.3	82.9	85.6	89.8	91.8	87.1	91.0
	2%	WB	79.8	78.7	78.2	76.1	73.1	70.6	71.0	71.4	72.6	76.2	75.8	78.9
		MCDB	89.7	87.5	88.2	85.3	81.5	77.5	79.3	83.1	86.6	88.3	86.9	89.6
	5%	WB	78.7	77.5	77.0	74.6	71.6	68.9	68.9	69.4	70.4	74.6	74.1	77.4
		MCDB	88.3	85.9	86.0	83.3	78.6	76.1	77.0	80.2	82.5	85.6	84.7	87.8
	10%	WB	77.5	76.4	75.6	73.4	70.0	67.3	67.1	67.3	68.6	72.8	72.7	76.1
		MCDB	86.8	84.6	83.8	81.5	76.0	74.0	74.7	77.2	78.8	81.9	81.5	85.9
Mean Daily Temperature Range	5% DB	MDBR	18.0	17.9	18.4	18.2	16.2	17.5	17.8	20.6	19.4	18.6	20.3	19.2
		MCDBR	21.6	21.7	21.9	20.8	20.0	19.5	20.7	24.6	24.6	24.5	25.9	23.5
		MCWBR	9.0	9.1	9.5	9.4	9.3	9.8	10.5	11.4	10.5	11.5	11.1	9.6
	5% WB	MCDBR	17.8	17.5	18.3	18.8	16.4	17.1	19.4	22.1	22.0	20.8	19.0	19.1
		MCWBR	9.1	8.5	9.5	10.1	8.9	10.4	11.4	12.2	10.8	11.1	10.2	9.5
Clear-Sky Solar Irradiance	taub	0.414	0.410	0.400	0.394	0.373	0.395	0.394	0.493	0.560	0.470	0.411	0.416	
	taud	2.392	2.403	2.418	2.406	2.430	2.367	2.345	2.018	1.860	2.173	2.351	2.366	
	Ebn at Noon	295	291	284	270	261	245	250	234	234	270	294	295	
	Edh at Noon	41	39	37	34	30	31	32	49	63	49	42	42	
All-Sky Solar Radiation	RadAvg	2088	1933	1680	1341	995	839	964	1166	1384	1656	2021	2118	
	RadStd	157	143	116	131	132	72	105	90	149	223	201	193	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+133

Nomenclature: See separate page